

SUPPLY CHAIN & LOGISTICS ANALYTICS

End-to-End Data Analytics Portfolio Project — Complete Project Workbook

Prepared as Internal Project Documentation

Data Analytics Team — Onboarding & Project Execution Guide

Field	Detail
Project Name	Supply Chain & Logistics Analytics
Project Type	Portfolio Project (Industry-Standard)
Industry Focus	Supply Chain, Warehouse, Inventory, Logistics, Procurement
Core Tools	Python, Pandas, NumPy, Matplotlib, PostgreSQL, Power BI, Git, GitHub
Dataset Format	Multiple relational CSV files
Document Type	Complete Project Workbook (Planning & Execution Guide)
Audience	Junior Data Analyst / New Team Member

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Section 1 — Project Overview

Project Objective

To design and build a complete, end-to-end Supply Chain & Logistics Analytics project that demonstrates the ability to collect, clean, model, analyze, and visualize multi-table operational data, and to translate that analysis into concrete business recommendations — replicating how a Data Analyst would operate inside a real supply chain organization.

Business Goal

Enable supply chain leadership to make faster, evidence-based decisions on inventory planning, warehouse efficiency, supplier reliability, shipment performance, and returns management by building a single analytics layer that connects customers, orders, products, suppliers, warehouses, inventory, shipments, and returns.

Expected Deliverables

Deliverable	Description
Cleaned Datasets	8 cleaned, validated, and standardized CSV/DB tables ready for analysis
PostgreSQL Database	Normalized relational schema with defined keys and relationships
EDA Notebooks	3 Jupyter notebooks covering cleaning, exploration, and business insights
SQL Query Bank	50 categorized business questions solved using SQL
Power BI Report	6 interconnected dashboards covering all operational areas
Business Insights Report	Executive summary with findings and recommendations
GitHub Repository	Portfolio-ready, professionally documented repository

Project Scope

- In Scope: Data cleaning, relational data modeling, exploratory data analysis, SQL-based business querying, Power BI dashboarding, and business recommendation reporting.
- In Scope: Analysis across customers, orders, products, suppliers, warehouses, inventory, shipments, and returns.
- Out of Scope: Real-time data pipelines, machine learning forecasting models, production deployment, and live system integration.
- Out of Scope: Financial audit-grade reconciliation; this is a decision-support analytics project, not a financial system of record.

Success Criteria

Criteria	Definition of Success
Data Quality	Zero critical nulls in key fields; duplicates resolved; consistent data types

Criteria	Definition of Success
	across all tables
Analytical Depth	Minimum 8 analytical themes covered with clear, quantified insights
SQL Proficiency	50 business questions answered using joins, subqueries, CTEs, and window functions
Dashboard Usability	6 dashboards, each answering its target audience's top 5 questions in under 3 clicks
Business Impact	Minimum 10 actionable, quantifiable recommendations delivered
Portfolio Readiness	Project structured, documented, and uploaded to GitHub in a professional format

Business Users

User Persona	Primary Interest
Supply Chain Director	High-level KPIs, cost trends, overall network health
Warehouse Operations Manager	Warehouse utilization, capacity, throughput
Procurement / Supplier Manager	Supplier reliability, lead times, quality
Logistics / Shipment Manager	On-time delivery, carrier performance, shipping cost
Customer Experience Manager	Returns, order fulfillment quality, customer satisfaction proxies
Finance / Business Analyst	Cost drivers, inventory carrying cost, ROI of operational changes

Project Assumptions

- The provided CSV files represent a realistic but simplified snapshot of a mid-size retail/e-commerce supply chain.
- Dates in all files follow a consistent, parseable format after cleaning.
- Each table contains a unique identifier suitable for use as a primary key.
- Currency values are assumed to be in a single consistent currency unless stated otherwise.
- The data does not require anonymization since it is synthetic/portfolio data.

Project Limitations

- No real-time or streaming data — all analysis is retrospective/historical.
- No external macroeconomic, weather, or fuel-price data is incorporated, which may otherwise explain shipment delays.
- Sample size and time range are limited to what is present in the CSV files.
- No forecasting or predictive modeling is included in this phase of the project.

Expected Business Value

- Reduced stockouts and overstocking through clearer inventory visibility.

- Improved on-time delivery performance through shipment and carrier analysis.
- Lower return rates through root-cause identification.
- Stronger supplier negotiations backed by data-driven performance scorecards.
- Better warehouse space and labor planning through utilization analysis.

Section 2 — Business Problem Statement

Current Situation

A growing multi-category retail company operates a network of regional warehouses, works with multiple suppliers, and ships products to customers across several regions. Operational data — customer orders, product catalogs, supplier records, warehouse inventories, shipment logs, and product returns — is captured, but it lives in disconnected files with no unified reporting layer. Decisions on stocking, warehousing, and supplier selection are currently made using manual spreadsheet reviews and inconsistent monthly summaries.

Problems

Problem Area	Description
Fragmented Data	Customer, order, product, supplier, warehouse, inventory, shipment, and return data are siloed across separate files with no relational model
Inventory Blind Spots	No consolidated view of stock levels vs. demand, leading to stockouts in some warehouses and overstock in others
Supplier Risk	No standardized way to compare supplier lead time, quality, and reliability
Shipment Delays	Delivery performance is not tracked systematically by carrier, region, or warehouse
Rising Returns	Return reasons and return rates by product/category are not analyzed, hiding quality or fulfillment issues
Slow Decision-Making	Manual reporting cycles delay operational and strategic decisions by weeks

Business Impact

- Lost sales from stockouts on high-demand products in specific warehouses.
- Increased holding costs from excess inventory of slow-moving products.
- Higher logistics costs from inefficient shipment routing and repeated late deliveries.
- Customer dissatisfaction and churn risk driven by delayed or incorrect deliveries and high return rates.
- Reduced negotiating leverage with underperforming suppliers due to lack of evidence.

Expected Outcome

A unified analytics solution — a normalized database, exploratory analysis, SQL query bank, and Power BI dashboard suite — that gives supply chain, warehouse, procurement, and logistics stakeholders a single source of truth to identify inefficiencies, quantify their cost, and act on prioritized, data-backed recommendations.

Success Metrics

Metric	Target / Direction
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Metric	Target / Direction
On-Time Delivery Rate	Increase; identify root causes of delays
Stockout Rate	Decrease through better inventory-demand alignment
Inventory Turnover	Increase for slow-moving SKUs identified
Return Rate	Decrease through root-cause targeting
Supplier On-Time Fulfillment	Increase via supplier scorecards
Warehouse Utilization Balance	Even out utilization across the network

Section 3 — Dataset Understanding

Fact Tables vs. Dimension Tables

The dataset follows a classic star-schema-style structure. Fact tables record transactional/event-level activity; dimension tables describe the entities involved.

Table	Type	Grain (One Row =)
orders.csv	Fact	One product line item within one customer order
shipments.csv	Fact	One shipment event for one order/order-item
inventory.csv	Fact (Snapshot)	Stock level of one product at one warehouse at a point in time
returns.csv	Fact	One returned item from one order
customers.csv	Dimension	One unique customer
products.csv	Dimension	One unique product/SKU
suppliers.csv	Dimension	One unique supplier
warehouses.csv	Dimension	One unique warehouse/location

Table Relationships (Conceptual)

- customers.csv (1) → orders.csv (Many): one customer places many orders.
- products.csv (1) → orders.csv (Many): one product appears in many order lines.
- suppliers.csv (1) → products.csv (Many): one supplier supplies many products.
- warehouses.csv (1) → inventory.csv (Many): one warehouse stores many product stock records.
- products.csv (1) → inventory.csv (Many): one product is stocked at many warehouses.
- orders.csv (1) → shipments.csv (Many/One): one order generates one or more shipments.
- orders.csv (1) → returns.csv (Many): one order can have one or more returned items.
- warehouses.csv (1) → shipments.csv (Many): one warehouse dispatches many shipments.

Primary Keys and Foreign Keys (Planning Reference)

Table	Primary Key	Foreign Keys
customers.csv	customer_id	—
products.csv	product_id	supplier_id → suppliers.csv
suppliers.csv	supplier_id	—
warehouses.csv	warehouse_id	—
orders.csv	order_id (+ order_item_id if line-level)	customer_id → customers.csv; product_id → products.csv; warehouse_id → warehouses.csv

Table	Primary Key	Foreign Keys
inventory.csv	inventory_id (or composite: product_id + warehouse_id)	product_id → products.csv; warehouse_id → warehouses.csv
shipments.csv	shipment_id	order_id → orders.csv; warehouse_id → warehouses.csv
returns.csv	return_id	order_id → orders.csv; product_id → products.csv

Data Dictionary — Column Categories

Columns across the eight files fall into the following functional categories. Use this to classify new columns discovered during initial inspection.

Category	Example Columns	Purpose
Identifiers	customer_id, order_id, product_id, supplier_id, warehouse_id, shipment_id, return_id	Uniquely identify records and link tables
Descriptive Attributes	customer_name, product_name, category, supplier_name, region, warehouse_location	Provide business context and labels for grouping
Dates / Time	order_date, ship_date, delivery_date, return_date	Enable trend, seasonality, and lag/duration analysis
Quantitative Measures	quantity, unit_price, stock_level, shipping_cost, order_value	Support aggregation, KPIs, and financial analysis
Categorical Flags / Status	order_status, shipment_status, return_reason, payment_mode	Segment data and calculate rates/percentages
Performance Metrics	lead_time_days, delivery_days, supplier_rating	Directly feed operational KPIs

Business Meaning of Every Dataset

customers.csv

Represents the people/organizations placing orders. Used to segment customers by region, order frequency, and value — supporting customer analysis and retention strategy.

products.csv

Represents the product catalog, including category and supplier linkage. Used to identify best-selling categories, slow movers, and pricing patterns.

suppliers.csv

Represents vendors who supply products. Used to evaluate lead time, quality/rating, and reliability, feeding supplier scorecards.

warehouses.csv

Represents physical storage/distribution locations. Used to analyze regional capacity, utilization, and dispatch efficiency.

inventory.csv

Represents stock levels of each product at each warehouse. Central to identifying stockouts, overstock, and reorder needs.

shipments.csv

Represents the fulfillment/delivery event for an order. Central to on-time delivery analysis, carrier/mode performance, and delay root-causing.

returns.csv

Represents products sent back by customers. Central to quality, fulfillment-accuracy, and customer-experience analysis.

orders.csv

Represents the transactional core of the business — what was bought, by whom, when, and for how much. Connects nearly every other table.

How the Datasets Interact With Each Other

An order (orders.csv) is placed by a customer (customers.csv) for one or more products (products.csv), which are sourced from suppliers (suppliers.csv) and stocked at warehouses (warehouses.csv) as tracked in inventory.csv. Once placed, the order is fulfilled via a shipment (shipments.csv) dispatched from a warehouse. If the customer is unsatisfied, the item may come back through returns.csv, referencing both the original order and product. This chain — Customer → Order → Product → Supplier/Warehouse → Shipment → Return — is the analytical backbone of the entire project.

Section 4 — Notebook Workbook

This section provides a complete, code-free workbook for each analytical notebook in the project. Each notebook entry is written so a junior analyst can follow it step-by-step without needing separate instructions.

Notebook 1 — 01_Data_Loading_&_Cleaning.ipynb

Notebook Objective

Load all eight raw CSV files, inspect their structure, and produce clean, validated, analysis-ready datasets.

Business Objective

Poor data quality upstream causes wrong insights downstream. This notebook protects every later decision by guaranteeing trustworthy data.

Task Checklist — Data Loading

- Import all 8 CSV files into separate DataFrames
- Confirm each file loaded with the expected number of rows/columns
- Standardize file/table naming conventions

Initial Inspection

- Check shape (rows, columns) of every table
- List column names and compare against the data dictionary
- Review data types for every column
- Identify missing values per column (count and percentage)
- Identify duplicate records per table
- Review unique value counts for key categorical columns

Cleaning Actions

- Convert all date columns to proper datetime format
- Standardize text casing (e.g., category names, statuses)
- Rename columns to consistent snake_case naming
- Handle missing values with a documented rule per column (drop, impute, or flag)
- Remove or investigate duplicate records
- Standardize categorical values (e.g., 'On-Time' vs 'on time' vs 'ONTIME')
- Validate numeric ranges (e.g., no negative quantities or prices)

Cleaning Validation

- Re-check shape, nulls, and duplicates after cleaning
- Confirm foreign key values in child tables exist in parent tables
- Spot-check 5–10 random records manually against source

Saving Clean Dataset

- Export each cleaned table to a dedicated 'clean data' folder

- Use a consistent file naming convention (e.g., clean_orders.csv)
- Log a short cleaning summary per table (rows before/after, nulls handled, duplicates removed)

Deliverables

- 8 cleaned CSV files ready for database loading and analysis
- A data cleaning log/summary documenting every transformation
- An updated data dictionary reflecting final column names and types

Expected Output

Eight clean, validated datasets with consistent types, no unresolved duplicates, standardized categorical values, and a documented cleaning log.

Professional Notes

- Treat this notebook as the foundation — every later insight depends on it.
- Never silently drop data; always document why a row or column was removed.
- Keep a copy of the raw, untouched files separate from the cleaned files.

Common Mistakes / Errors

- Assuming column names are already clean without checking for hidden whitespace or casing issues.
- Converting dates without checking for multiple formats mixed in the same column.
- Removing duplicates without first confirming they are true duplicates and not valid repeat transactions.
- Forgetting to re-validate data after cleaning steps are applied.

Professional Tips

- Build a small reusable cleaning checklist/function you re-run after every transformation.
- Keep a 'data quality scorecard' summarizing null %, duplicate %, and type issues per table.
- Cross-check record counts between related tables (e.g., every order_id in shipments exists in orders).

Notebook 2 — 02_Exploratory_Data_Analysis.ipynb

Notebook Objective

Explore every dataset individually and jointly to uncover patterns, relationships, and operational signals across the supply chain.

Business Objective

Surface the patterns that justify the business recommendations delivered later — this notebook is the evidence base for every insight in Notebook 3.

Overall Dataset Analysis

- Summary statistics for all numeric fields across all 8 tables
- Record counts and time range covered by the data
- High-level volume trends (orders, shipments, returns over time)

Customer Analysis

- Customer distribution by region
- Order frequency per customer (one-time vs. repeat)
- Customer value distribution (total spend per customer)
- Top 10 customers by revenue

Supplier Analysis

- Number of products per supplier
- Average supplier rating distribution
- Supplier lead time comparison

Warehouse Analysis

- Number of products and orders handled per warehouse
- Regional distribution of warehouses
- Warehouse-level shipment volume

Inventory Analysis

- Current stock level distribution by product/category
- Low-stock / stockout candidates (below reorder threshold)
- Overstocked products by warehouse

Shipment Analysis

- On-time vs. delayed shipment ratio
- Average delivery time by warehouse and by region
- Shipping cost distribution by mode/carrier

Returns Analysis

- Return rate overall and by category
- Top return reasons
- Return rate by warehouse and by supplier-linked product

Regional & Category Analysis

- Revenue and order volume by region
- Revenue and return rate by product category

Monthly Trends & Seasonality

- Monthly order volume and revenue trend
- Seasonal peaks (e.g., holiday months) in orders and returns
- Month-over-month growth rate

Product, Segment & Mode Analysis

- Best-selling and worst-selling products
- Customer segments by spend/frequency (e.g., high/medium/low value)
- Shipping mode usage and associated cost/speed trade-off
- Warehouse utilization rate (stock held vs. capacity/throughput proxy)
- Supplier rating vs. lead time vs. return rate correlation view

Operational KPIs & Performance Metrics

- On-time delivery rate

- Average order fulfillment time
- Inventory turnover proxy
- Return rate %
- Average supplier lead time
- Revenue per warehouse / per region

Deliverables

- A structured EDA notebook covering all 8 analytical themes above
- A set of annotated charts answering each business question
- A running list of 'insight candidates' to carry into Notebook 3

Expected Output

A complete visual and statistical exploration of the supply chain, with every chart tied to a specific business question and a documented takeaway.

Professional Notes

- State the business question before each chart — never plot without a purpose.
- Annotate every chart with a one-line takeaway, not just a title.
- Compare metrics across at least two dimensions (e.g., region AND category) to avoid shallow conclusions.

Common Mistakes / Errors

- Producing charts with no business question attached ('chart for chart's sake').
- Drawing conclusions from a single aggregate view without checking segment-level differences.
- Ignoring outliers instead of investigating whether they represent real operational issues.
- Confusing correlation (e.g., rating vs. returns) with causation.

Professional Tips

- Build a 'Business Questions Answered' table at the top of the notebook and check items off as you go.
- Use consistent color themes for charts across the whole notebook for a professional look.
- Save key charts as images for direct reuse in the Business Insights notebook and Power BI report.

Notebook 2 — Business Questions Bank

Theme	Sample Business Question	Chart Type
Customer	Which regions generate the most revenue and orders?	Bar chart by region
Supplier	Which suppliers have the longest lead times?	Horizontal bar / box plot
Warehouse	Which warehouses are over- or under-utilized?	Bar chart with threshold line
Inventory	Which products are at risk of stockout this month?	Ranked table / bar chart
Shipment	What % of shipments are delivered on time by region?	Stacked bar chart
Returns	What are the top 3 reasons for returns by category?	Pareto chart
Trend	How does order volume change month over month?	Line chart

Theme	Sample Business Question	Chart Type
Category	Which categories have the highest return rate?	Bar chart

Expected Insights (Notebook 2)

- A ranked list of the top and bottom performing regions, warehouses, and suppliers.
- Clear identification of products/categories driving the highest return rates.
- A quantified on-time delivery rate broken down by warehouse and shipping mode.
- Seasonal patterns in demand that inventory planning should account for.

Notebook 3 — 03_Business_Insights.ipynb

Notebook Objective

Consolidate the findings from Notebook 2 into an executive-ready narrative of insights, risks, and recommendations.

Business Objective

Translate analysis into decisions — this notebook is what a manager or director will actually read.

Executive Summary

- 3–5 sentence overview of the project and headline findings
- Top 5 key numbers a director should know

Key Findings by Area

- Customer Insights: who buys, where, and how much
- Warehouse Insights: capacity and utilization imbalances
- Inventory Insights: stockout and overstock risk products
- Supplier Insights: reliability and lead-time performance
- Shipping Insights: on-time performance and cost drivers
- Returns Insights: root causes and highest-impact categories

Opportunity & Risk Identification

- Cost reduction opportunities (e.g., overstock carrying cost, expensive shipping modes)
- Risk areas (e.g., single supplier dependency, chronic warehouse delays)
- Operational bottlenecks (e.g., slowest link in the fulfillment chain)
- Improvement opportunities ranked by estimated impact vs. effort

Deliverables

- A written Executive Summary (1 page)
- A structured Key Findings section per operational area
- A prioritized Recommendations list ready for the Business Recommendations section

Expected Output

A polished, narrative business insights document that can be shared directly with supply chain leadership.

Professional Notes

- Write for a non-technical audience — translate statistics into business language.
- Every finding should answer 'so what' — why does this matter to the business.
- Tie every recommendation back to a specific, quantified finding.

Common Mistakes / Errors

- Repeating charts from Notebook 2 without synthesizing them into a narrative.
- Making recommendations that are not clearly supported by the data shown.
- Overloading the summary with numbers instead of a clear story.

Professional Tips

- Use the 'Situation → Insight → Recommendation' structure for each finding.
- Rank recommendations by business impact and ease of implementation.
- Keep the Executive Summary to one page — leadership reads that first (and sometimes only).

Section 5 — SQL Workbook

This section is a planning roadmap for the database and query layer of the project. No SQL code is included — only structure, sequencing, and the business question bank.

Database Planning

- Create one PostgreSQL database dedicated to this project (e.g., supply_chain_analytics).
- Create one schema (e.g., public or a named schema like scm) to hold all 8 tables.
- Plan table creation order based on dependency: dimension tables first, fact tables second.
- Reserve a staging area (or staging schema) for raw imports before applying constraints.

Normalization

- Target Third Normal Form (3NF) for dimension tables (customers, products, suppliers, warehouses).
- Keep fact tables (orders, shipments, inventory, returns) normalized but optimized for join performance.
- Avoid repeating descriptive attributes (e.g., customer name) inside fact tables — reference by ID instead.
- Document any deliberate denormalization decisions made for reporting convenience.

ER Diagram Planning

- Plan a star-schema-like ER diagram: orders, shipments, inventory, and returns as fact tables radiating from customers, products, suppliers, and warehouses as dimensions.
- Diagram should show cardinality (1:1, 1:Many) on every relationship.
- Diagram should visually separate dimension tables (top/sides) from fact tables (center).

Table Relationships, Primary Keys & Foreign Keys

Refer to Section 3 for the full primary key / foreign key reference table. Use it directly when creating the schema.

Import Workflow

- Import cleaned CSVs (output of Notebook 1) into staging tables.
- Validate row counts against the cleaned CSV files after import.
- Apply primary key and foreign key constraints only after validating referential integrity.
- Move validated data from staging into final production tables.

Validation Workflow

- Check for orphaned foreign keys (e.g., orders referencing a non-existent customer_id).
- Reconcile row counts between the CSV source and the database table.
- Spot-check aggregate totals (e.g., total order count, total revenue) against the notebook EDA results.

Business KPIs to Plan For

KPI	Business Meaning
On-Time Delivery Rate	% of shipments delivered on or before the expected date
Average Lead Time	Average days from order to fulfillment/supplier delivery
Inventory Turnover	Rate at which stock is sold and replaced over a period
Return Rate	% of ordered items returned
Revenue per Region/Warehouse	Total order value grouped by region or warehouse
Supplier Reliability Score	Composite of lead time consistency and product quality/rating

Query Roadmap

Technique	Where It Will Be Used
Aggregation (GROUP BY, SUM, AVG, COUNT)	Revenue by region, average lead time by supplier, return rate by category
JOINS (INNER, LEFT)	Combining orders with customers/products; shipments with warehouses
Subqueries	Finding customers above average spend; products below average stock
CTEs (WITH clauses)	Multi-step KPI calculations such as monthly revenue growth
Window Functions	Running totals, rankings (top N per category), month-over-month comparisons
Views	Reusable views for 'on-time shipments', 'active low-stock products', 'top customers'
Indexes	On foreign key columns and frequently filtered columns (order_date, region, category)
Performance Optimization	Review query plans (EXPLAIN ANALYZE) for slow aggregate queries on large joins

SQL Documentation Structure

- sql/01_schema/ — table creation scripts (planning notes only here, per project rules)
- sql/02_constraints/ — primary/foreign key and validation notes
- sql/03_business_questions/ — one file per difficulty tier (easy, medium, advanced, expert)
- sql/04_views/ — planned reusable views
- sql/05_performance/ — indexing and optimization notes

50 SQL Business Questions

Categorized by difficulty. These define the scope of SQL practice and the analytical questions the database must be able to answer. No SQL code is provided — only the questions.

Easy (1–15)

1. How many total customers are in the dataset?
2. How many total orders have been placed?

3. List all products belonging to a specific category.
4. What is the total revenue generated from all orders?
5. How many suppliers does the company work with?
6. How many warehouses are currently in operation?
7. List all orders placed in a specific month.
8. What is the average unit price of all products?
9. How many shipments were marked as delayed?
10. How many products does each supplier provide?
11. What is the total quantity of each product currently in stock?
12. List all customers from a specific region.
13. How many returns have been recorded in total?
14. What is the average shipping cost across all shipments?
15. List all distinct shipping modes used.

Medium (16–30)

16. What is the total revenue generated per region?
17. Which product category has the highest number of returns?
18. What is the average lead time per supplier?
19. Which warehouse handles the highest volume of shipments?
20. What is the monthly order trend over the last 12 months?
21. Which customers have placed more than 5 orders?
22. What is the on-time delivery rate by warehouse?
23. Which products have stock levels below a defined reorder threshold?
24. What is the average order value per customer segment?
25. Which supplier has the highest average product rating?
26. What percentage of orders result in a return?
27. Which shipping mode has the highest average delivery delay?
28. What is the total inventory value held per warehouse?
29. Which region has the highest return rate?
30. What is the year-over-year growth in total revenue?

Advanced (31–42)

31. Using a CTE, calculate month-over-month revenue growth percentage.
32. Rank suppliers by average lead time within each product category using a window function.
33. Identify the top 3 best-selling products per category using a ranking window function.
34. Calculate a running total of monthly revenue across the year.
35. Find customers whose total spend is above the overall average spend using a subquery.
36. Identify products that are in the bottom 10% of sales but top 10% of returns.
37. Calculate the month-over-month change in on-time delivery rate.
38. Identify warehouses whose average shipment delay increased for three consecutive months.
39. Find the top 5 customers by revenue in each region using a window function.
40. Calculate each supplier's return rate relative to the company-wide average return rate.

41. Identify products with consistently declining monthly sales over the last 6 months.
42. Calculate the percentage contribution of each category to total company revenue using a CTE.

Expert (43–50)

43. Build a query that flags 'at-risk' suppliers based on a composite score of lead time trend, rating, and return rate.
44. Create a single query that reconciles inventory levels against unfulfilled order demand per warehouse.
45. Design a query to identify cascading delay effects — orders delayed due to shipments delayed due to inventory shortages.
46. Build a cohort-style query analyzing repeat return behavior by customer over time.
47. Construct a query that estimates lost revenue from stockout events using order and inventory history.
48. Build a multi-CTE query producing a full supplier scorecard (lead time, rating, return rate, revenue contribution).
49. Design a query identifying warehouses that are simultaneously over capacity and under-delivering on time.
50. Create a query that segments customers into value tiers (using window functions) and reports return behavior per tier.

Section 6 — Power BI Workbook

This section plans the full Power BI report suite. Six dashboards cover every operational area, each designed for a specific business audience identified in Section 1.

Dashboard 1 — Executive Dashboard

Objective

Give supply chain leadership a single-page, high-level view of overall network health, revenue, and performance trends.

Target Users

- Supply Chain Director
- VP Operations
- Finance Leadership

KPIs

- Total Revenue
- Total Orders
- On-Time Delivery Rate
- Overall Return Rate
- Average Order Value
- Revenue Growth %

Charts

- Monthly revenue trend line chart
- Revenue by region map/bar chart
- Order volume trend line chart
- Category revenue contribution donut/bar chart

Cards

- Total Revenue card
- Total Orders card
- On-Time Delivery % card
- Return Rate % card

Filters

- Date range filter
- Region filter
- Category filter

Slicers

- Region slicer
- Time period slicer (month/quarter/year)

Bookmarks / Tooltips / Drill Through / Navigation

- Bookmark to switch between 'Revenue View' and 'Operations View'
- Tooltip showing prior-period comparison on hover
- Drill through to Warehouse/Supplier/Shipment dashboards from any KPI card
- Top navigation bar linking to all 6 dashboards

Business Questions Answered

- What is our overall revenue and order trend?
- Which regions are outperforming or underperforming?
- Are we hitting our on-time delivery targets company-wide?

Insights

- Revenue growth is concentrated in a subset of regions/categories.
- On-time delivery has a clear seasonal dip during peak order months.

Recommendations

- Prioritize investment in high-growth regions.
- Set a formal on-time delivery target and review it monthly at the executive level.

Expected Outcome

A single, glanceable landing page that answers 80% of leadership's recurring questions in under 10 seconds.

Dashboard 2 — Warehouse Dashboard

Objective

Help warehouse operations managers monitor capacity, utilization, and shipment throughput per location.

Target Users

- Warehouse Operations Manager
- Regional Operations Lead

KPIs

- Warehouse Utilization %
- Total Shipments Dispatched
- Average Dispatch Time
- Stock Held per Warehouse

Charts

- Utilization by warehouse bar chart
- Shipment volume by warehouse over time line chart
- Warehouse capacity vs. stock held comparison chart

Cards

- Total Warehouses card
- Average Utilization % card
- Total Stock Value card

Filters

- Warehouse filter
- Region filter

- Date range filter

Slicers

- Warehouse slicer
- Region slicer

Bookmarks / Tooltips / Drill Through / Navigation

- Drill through from a warehouse bar to that warehouse's product-level inventory detail
- Tooltip showing warehouse address/region on hover
- Bookmark for 'Overstocked View' vs 'Understocked View'

Business Questions Answered

- Which warehouses are over- or under-utilized?
- Which warehouses have the highest dispatch volume?
- Where is stock imbalanced relative to demand?

Insights

- A small number of warehouses handle a disproportionate share of shipment volume.
- Certain regional warehouses are consistently near capacity while others are underused.

Recommendations

- Rebalance stock allocation from over-utilized to under-utilized warehouses.
- Evaluate adding capacity or shifting demand for the most strained locations.

Expected Outcome

Clear visibility into which warehouses need operational or capacity intervention.

Dashboard 3 — Inventory Dashboard

Objective

Give inventory planners a real-time view of stock health, stockout risk, and overstock exposure.

Target Users

- Inventory Planner
- Procurement Manager
- Warehouse Manager

KPIs

- Total Stock Units
- Stockout Risk Count
- Overstock Count
- Inventory Turnover Proxy

Charts

- Stock level by category bar chart
- Low-stock products ranked table/chart
- Stock distribution by warehouse heatmap-style matrix

Cards

- Total SKUs card

- At-Risk SKUs card
- Overstocked SKUs card

Filters

- Category filter
- Warehouse filter
- Stock status filter (low/normal/high)

Slicers

- Category slicer
- Warehouse slicer

Bookmarks / Tooltips / Drill Through / Navigation

- Drill through from a low-stock product to its full order/shipment history
- Tooltip showing days-of-stock-remaining estimate
- Bookmark for 'Stockout Risk View' vs 'Overstock View'

Business Questions Answered

- Which products are at risk of stocking out?
- Which products are overstocked and tying up capital?
- How is stock distributed across warehouses for a given product?

Insights

- A specific set of high-demand products is repeatedly at stockout risk in the same warehouses.
- Slow-moving products are concentrated in a few categories, tying up warehouse space.

Recommendations

- Set automated reorder points for the identified at-risk SKUs.
- Run a clearance or redistribution plan for overstocked slow-moving products.

Expected Outcome

A proactive early-warning view that lets planners act before stockouts or overstock become costly.

Dashboard 4 — Supplier Dashboard

Objective

Help procurement evaluate and compare supplier performance for negotiation and sourcing decisions.

Target Users

- Procurement Manager
- Supplier Relationship Manager

KPIs

- Average Lead Time
- Average Supplier Rating
- Number of Products per Supplier
- Supplier Return Rate

Charts

- Lead time by supplier bar chart

- Supplier rating vs. lead time scatter chart
- Return rate by supplier bar chart

Cards

- Total Suppliers card
- Average Lead Time card
- Top-Rated Supplier card

Filters

- Supplier filter
- Category filter
- Date range filter

Slicers

- Supplier slicer
- Category slicer

Bookmarks / Tooltips / Drill Through / Navigation

- Drill through from a supplier to its full product list and performance history
- Tooltip showing supplier contract/region details on hover
- Bookmark for 'Top Performers' vs 'At-Risk Suppliers' view

Business Questions Answered

- Which suppliers have the longest lead times?
- Which suppliers have the highest product return rates?
- Which suppliers should we consider renegotiating with or replacing?

Insights

- A handful of suppliers account for a disproportionate share of long lead times and returns.
- Higher-rated suppliers correlate with lower return rates, supporting a quality-first sourcing strategy.

Recommendations

- Open renegotiation or performance-improvement conversations with the lowest-scoring suppliers.
- Shift volume gradually toward consistently high-performing suppliers.

Expected Outcome

A supplier scorecard view that gives procurement clear, evidence-based leverage in vendor conversations.

Dashboard 5 — Shipment Dashboard

Objective

Help logistics managers monitor delivery performance, delays, and shipping cost efficiency.

Target Users

- Logistics Manager
- Shipment/Carrier Coordinator

KPIs

- On-Time Delivery Rate
- Average Delivery Time

- Total Shipping Cost
- Delayed Shipment Count

Charts

- On-time vs. delayed shipments by month line/bar chart
- Delivery time by shipping mode bar chart
- Shipping cost by region/warehouse chart

Cards

- On-Time Delivery % card
- Average Delivery Days card
- Total Shipping Cost card

Filters

- Shipping mode filter
- Warehouse filter
- Date range filter

Slicers

- Shipping mode slicer
- Region slicer

Bookmarks / Tooltips / Drill Through / Navigation

- Drill through from a delayed-shipment bar to the underlying order/warehouse detail
- Tooltip showing carrier/mode cost-vs-speed trade-off
- Bookmark for 'On-Time View' vs 'Delay Root-Cause View'

Business Questions Answered

- What is our on-time delivery rate by region and mode?
- Which shipping modes are slowest or most expensive?
- Which warehouses generate the most delayed shipments?

Insights

- One or two shipping modes account for a disproportionate share of delays.
- Delivery delays are concentrated around specific warehouses or regions, suggesting localized bottlenecks.

Recommendations

- Re-evaluate carrier/mode selection for the routes with the worst delay rates.
- Investigate operational bottlenecks at the warehouses driving the most delays.

Expected Outcome

A performance monitoring view that helps logistics teams catch and address delay patterns quickly.

Dashboard 6 — Returns Dashboard

Objective

Help customer experience and quality teams understand why products are returned and where to intervene.

Target Users

- Customer Experience Manager
- Quality/Product Manager

KPIs

- Total Returns
- Return Rate %
- Top Return Reason
- Return Rate by Category

Charts

- Return reason breakdown bar/Pareto chart
- Return rate trend over time line chart
- Return rate by category and warehouse chart

Cards

- Total Returns card
- Return Rate % card
- Top Return Reason card

Filters

- Category filter
- Return reason filter
- Date range filter

Slicers

- Category slicer
- Region slicer

Bookmarks / Tooltips / Drill Through / Navigation

- Drill through from a return reason to the underlying product/order detail
- Tooltip showing customer segment on hover
- Bookmark for 'By Category' vs 'By Reason' view

Business Questions Answered

- Why are customers returning products?
- Which categories/products have the highest return rates?
- Is the return rate improving or worsening over time?

Insights

- A small set of return reasons accounts for the majority of returns (Pareto pattern).
- Return rate is notably higher for specific categories, pointing to quality or description-accuracy issues.

Recommendations

- Address the top 1–2 return reasons directly (e.g., sizing/description accuracy, packaging damage).
- Flag high-return products to the supplier/quality team for review.

Expected Outcome

A root-cause view of returns that connects customer experience directly to sourcing and fulfillment quality.

Section 7 — Business Recommendations

Recommendations are organized by time horizon and by functional area so they can be picked up directly by different teams.

Short-Term Recommendations (0–3 Months)

- Set automated reorder alerts for the SKUs identified as chronic stockout risks.
- Redistribute overstocked inventory from low-utilization warehouses to high-demand ones.
- Open performance review conversations with the lowest-scoring suppliers on lead time and rating.
- Address the single highest-volume return reason with a quick-win fix (e.g., improved product description or packaging).

Medium-Term Recommendations (3–9 Months)

- Introduce a formal supplier scorecard reviewed quarterly, combining lead time, rating, and return rate.
- Rebalance warehouse capacity allocation based on the utilization analysis from Dashboard 2.
- Pilot a change in shipping mode/carrier for the routes with the worst on-time delivery performance.
- Build a standing monthly operations review using the Power BI report suite as the single source of truth.

Long-Term Recommendations (9+ Months)

- Evaluate diversifying suppliers for product categories currently dependent on a single low-rated vendor.
- Invest in warehouse capacity expansion or automation for consistently over-utilized locations.
- Explore predictive inventory planning (forecasting) as a follow-on project once this analytics foundation is in place.
- Formalize a company-wide returns-reduction program tied to the top return-reason categories.

Focused Strategy Areas

Area	Strategy
Cost Reduction	Target the highest-cost, lowest-performing shipping modes and the most expensive overstocked SKUs first
Inventory Optimization	Set data-backed reorder points and safety stock levels per product-warehouse combination
Warehouse Optimization	Rebalance stock and shipment load across warehouses based on utilization data
Supplier Improvement	Move volume toward high-rating, low-return, on-time suppliers; formalize scorecards
Customer Experience	Reduce returns by addressing the top 1–2 root causes identified in the Returns Dashboard
Risk Mitigation	Diversify supplier base for high-dependency categories; monitor warehouses with rising delay trends

Section 8 — Project Documentation Checklist

Use this checklist before considering the project portfolio-ready and before uploading to GitHub.

Folder Structure

- ☐ Root project folder created with a clear, professional name
- ☐ Separate folders for: data/, notebooks/, sql/, powerbi/, images/, docs/
- ☐ data/ split into data/raw/ and data/clean/

Notebook Structure

- ☐ 01_Data_Loading_&_Cleaning.ipynb present and runs top-to-bottom without errors
- ☐ 02_Exploratory_Data_Analysis.ipynb present with all charts rendering correctly
- ☐ 03_Business_Insights.ipynb present with a clear written narrative
- ☐ Notebooks numbered and named consistently

SQL Folder

- ☐ Schema/creation scripts organized by table
- ☐ Business question queries organized by difficulty tier (easy/medium/advanced/expert)
- ☐ Views and indexing scripts documented separately

Power BI Folder

- ☐ .pbix file saved and named clearly
- ☐ All 6 dashboards present and functioning with working filters/slicers
- ☐ Data model relationships verified (no broken/ambiguous joins)

Images, Dataset, and Report Folders

- ☐ images/ folder contains exported chart and dashboard screenshots
- ☐ Original raw datasets preserved and clearly labeled as 'raw' (unmodified)
- ☐ ER diagram image included in docs/
- ☐ Data dictionary document included in docs/
- ☐ Final project report (PDF or doc) summarizing the whole project included

README & GitHub Repository

- ☐ README includes project overview, problem statement, tools used, and key findings
- ☐ README includes screenshots of at least 2 dashboards
- ☐ Repository has a clear commit history reflecting project stages (not one giant commit)
- ☐ .gitignore excludes unnecessary files (e.g., checkpoints, temp files)
- ☐ Repository is public and the link is tested in an incognito browser

Portfolio Readiness Checklist

Check	Status Question
Clarity	Can a stranger understand the business problem in under 2 minutes?
Completeness	Are all 8 project sections (overview through roadmap) represented in the repo?
Visual Quality	Do all charts and dashboards have titles, labels, and a consistent color theme?
Narrative	Does the project tell a clear story from problem → analysis → recommendation?
Professionalism	Is naming, folder structure, and grammar consistent throughout?

Section 9 — Interview Preparation

60 interview questions across 10 categories, 6 each. Answers are intentionally not included — this is a question bank to prepare against, not a script.

1. Project Understanding

1. Walk me through this project end-to-end in two minutes.
2. Why did you choose a supply chain and logistics dataset for this project?
3. What was the most challenging part of this project?
4. How are the 8 datasets in this project related to each other?
5. What tools did you use in each stage of the project, and why?
6. If you had one more week, what would you add to this project?

2. Business Understanding

7. What business problem does this project actually solve?
8. Who would use the dashboards you built, and what decisions would they make from them?
9. How would you explain 'inventory turnover' to a non-technical stakeholder?
10. Why does on-time delivery rate matter to a business beyond customer satisfaction?
11. How do supplier performance and return rate connect from a business perspective?
12. What is the business cost of a stockout versus an overstock?

3. Python

13. How did you handle missing values in this project, and why did you choose that method?
14. What is the difference between a list, tuple, and dictionary, and where might each appear in this project?
15. How would you detect and remove duplicate records in a dataset?
16. What is the difference between loc and iloc in Pandas?
17. How do you convert a column to datetime format, and why is that necessary?
18. What Python libraries did you use in this project, and what is each one responsible for?

4. Pandas

19. How would you merge two DataFrames on a common key, and what join types are available?
20. What is the difference between groupby() and pivot_table()?
21. How do you identify and handle outliers in a numeric column?
22. What does apply() do, and when would you use it over a vectorized operation?
23. How would you calculate a rolling average or month-over-month change in Pandas?
24. How do you check and fix data type mismatches across multiple related DataFrames?

5. Exploratory Data Analysis (EDA)

25. What was your process for deciding which charts to create in this project?
26. How do you decide whether a pattern in the data is meaningful or just noise?
27. What is the difference between univariate, bivariate, and multivariate analysis?

- 28. How did you analyze seasonality in the order/shipment data?
- 29. What would make you suspect that a correlation you found is misleading?
- 30. How do you decide which KPIs matter most for a supply chain dataset?

6. SQL

- 31. What is the difference between an INNER JOIN and a LEFT JOIN, and when would you use each?
- 32. How would you find the top 3 customers by revenue in each region using SQL?
- 33. What is a CTE, and why would you use one over a subquery?
- 34. Explain how a window function like RANK() differs from GROUP BY.
- 35. How would you write a query to calculate month-over-month revenue growth?
- 36. How would you find orders that don't have a matching shipment record?

7. Power BI

- 37. How did you decide what should go on the Executive Dashboard versus the operational dashboards?
- 38. What is the difference between a slicer and a filter in Power BI?
- 39. How do drill-through and bookmarks improve a dashboard's usability?
- 40. How would you design a dashboard for a warehouse manager differently than one for a CEO?
- 41. What is the difference between a calculated column and a measure in Power BI?
- 42. How would you handle a data model with ambiguous relationships between tables?

8. Business Insights

- 43. What was the single most important insight from this project, and why?
- 44. How did you prioritize which recommendations to make first?
- 45. How would you measure whether your recommendations actually worked after implementation?
- 46. What insight surprised you the most during this analysis?
- 47. How do you avoid presenting too many insights and losing the audience's attention?
- 48. If leadership could only act on one recommendation, which would you tell them to pick, and why?

9. Scenario-Based

- 49. A warehouse manager disagrees with your utilization numbers — how do you respond?
- 50. You discover a data quality issue after the dashboard has already been shared — what do you do?
- 51. Leadership wants a single number to represent 'supply chain health' — how would you approach building that?
- 52. A supplier scorecard shows a top supplier suddenly dropping in performance — how would you investigate?
- 53. You have conflicting priorities: reduce returns vs. reduce shipping cost — how do you decide what to recommend?
- 54. Your dashboard refresh is taking too long for daily use — how would you approach optimizing it?

10. HR / Behavioral

- 55. Tell me about yourself.
- 56. Why do you want to work in a Data Analyst / Business Development role?
- 57. Describe a time you had to explain a technical result to a non-technical person.

- 58. How do you handle feedback or criticism on your analysis?
- 59. Where do you see yourself in three years?
- 60. Why should we hire you for this role?

Section 10 — Final Project Roadmap

A day-by-day / phase-by-phase roadmap from project kickoff to final GitHub upload. Adjust durations to your own pace, but keep the sequence — each phase depends on the one before it.

Phase	Days	Focus	Key Output
Phase 1: Setup & Understanding	Day 1–2	Read this workbook fully; set up folder structure, Git repo, and environment; review all 8 raw CSV files	Project skeleton + initial data familiarity
Phase 2: Data Loading & Cleaning	Day 3–5	Complete Notebook 1: load, inspect, clean, validate, and export all 8 tables	8 clean datasets + cleaning log
Phase 3: Database Setup	Day 6–7	Design ER diagram; create PostgreSQL schema; import cleaned data; validate referential integrity	Working normalized database
Phase 4: Exploratory Data Analysis	Day 8–12	Complete Notebook 2: run all analytical themes, produce and annotate all required charts	Full EDA notebook with insight candidates
Phase 5: SQL Business Questions	Day 13–16	Answer all 50 categorized business questions using joins, subqueries, CTEs, and window functions	Completed SQL query bank
Phase 6: Business Insights Notebook	Day 17–18	Complete Notebook 3: executive summary, key findings, opportunities, and risks	Business insights narrative document
Phase 7: Power BI Dashboard Build	Day 19–24	Build the data model and all 6 dashboards with KPIs, charts, filters, and navigation	Complete .pbix report suite
Phase 8: Business Recommendations	Day 25	Consolidate short/medium/long-term recommendations by focus area	Business recommendations document
Phase 9: Documentation & Polish	Day 26–27	Finalize README, data dictionary, ER diagram image, and folder structure per the Section 8 checklist	Portfolio-ready documentation
Phase 10: GitHub Upload & Review	Day 28	Upload repository, verify all links and images render, do a final end-to-end review as if you were a hiring manager	Live, portfolio-ready GitHub repository

Final Pre-Submission Review Checklist

Before considering the project complete, confirm every item in Section 8's Portfolio Readiness Checklist is satisfied, all three notebooks run cleanly end-to-end, all 6 Power BI dashboards load without errors, and the GitHub README accurately reflects the finished project.